Temporal and Transverse Hanbury Brown–Twiss Constraints on Spacetime Foam

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We present a dimensionally consistent framework for testing spacetime foam models characterized by a single power-law parameter α using temporal (single-telescope) Hanbury Brown–Twiss (HBT) intensity-correlation measurements. We extend an earlier line-of-sight (1D) formalism by incorporating both temporal and transverse foam fluctuations across an extended source. Foam induces random photon arrival-time jitter, shifting detected arrival times by $\delta t = \Delta\ell_{\rm tot}^{(D)}/c$. The arrival-time jitter becomes ray-dependent for photons arriving from different source directions. We compute the variance of the differential arrival-time increment between two rays as

$$\sigma_\Delta^2(\tau;\boldsymbol{\theta},\boldsymbol{\theta}') \;=\; \frac{2\sigma_\ell^2(D)}{c^2}\Big[1 - C_\alpha(\tau;D)\,\Phi\big(b(\boldsymbol{\theta},\boldsymbol{\theta}')/r_\perp\big)\Big],$$

where C_α encodes temporal decorrelation and Φ encodes transverse decorrelation with transverse correlation length $r_\perp$. The observed second-order correlation is an angularly weighted (extended-source) mixture of Gaussian-jitter convolutions of the intrinsic source correlation. In the limit of large transverse foam coherence across the source (point-source/fully correlated regime), the prediction reduces to the earlier 1D result. When transverse foam coherence is small, extended sources experience enhanced effective jitter, broadening and suppressing the bunching peak. Pure-phase, strictly single-mode modulation leaves $g^{(2)}(\tau)$ unchanged; observable effects require arrival-time jitter (or a mechanism that couples phase to amplitude).

I. INTRODUCTION

Spacetime foam, originally proposed by Wheeler [1] and developed in subsequent quantum-gravity work [2, 3], refers to hypothesized granularity of spacetime at scales near the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. A phenomenological framework by Ng and collaborators [4–6] parametrizes a broad class of foam models by a single power-law exponent α controlling how distance fluctuations accumulate with propagation length. For a photon propagating a distance ℓ,

$$\delta\ell \;\sim\; \ell^{1-\alpha}\,\ell_{\rm P}^{\alpha}, \tag{1}$$

with conservative fluctuations corresponding to $\alpha = 1$ (giving $\delta\ell \sim \ell_{\rm P}$ independent of ℓ), holography-inspired models to $\alpha = 2/3$, and random-walk models to $\alpha = 1/2$.

Lieu and Hillman [7] proposed constraining spacetime foam through loss of phase coherence in light from distant extragalactic sources, while subsequent work [4, 8, 9] developed related phase-coherence and imaging constraints. Carlip [10] cautioned that some constraints may inherit dimensional ambiguities from assumptions about what kinds of metric fluctuations dominate observable signatures and what statistical properties/correlation structures are appropriate.

Hanbury Brown–Twiss (HBT) intensity correlations offer an alternative temporal-coherence probe. Temporal (single-telescope) HBT measures the second-order intensity correlation $g^{(2)}(\tau)$ at a single spatial location as a function of time lag τ, directly probing temporal coherence. Modern HBT campaigns targeting bright stars and active galactic nuclei (AGN) have begun achieving photon-rate sensitivity adequate for precision tests [11, 12].

An earlier line-of-sight (1D) formulation emphasized that a strictly pure-phase, single-mode foam does not alter $g^{(2)}(\tau)$; observable HBT effects require arrival-time jitter rather than phase decoherence. Here we extend the formalism to include transverse foam fluctuations across an extended source and show explicitly how the foam correlation length in the transverse direction controls whether the point-source approximation is valid. If the transverse coherence of the foam is large compared to the source's angular extent (across the rays contributing to detection), the extended source effectively behaves point-like for the foam-induced arrival-time jitter mechanism. If the transverse coherence is small, photons arriving along different lines of sight receive effectively independent timing jitter, increasing the effective decoherence and further suppressing/broadening the HBT bunching peak.

The paper is organized as follows. Section II establishes notation, dimensional analysis, and the ray/angle representation for extended sources. Section III derives temporal and transverse foam statistics and their implications

for differential arrival-time jitter. Section IV connects the foam-induced jitter to the HBT observable, including the key pure-phase cancellation. Section V provides the explicit angularly averaged convolution formula for $g^{(2)}_{\rm obs}(\tau)$. Section VI gives a parameterized numerical illustration including a transverse-coherence factor. Finally, Section VII discusses limitations and next steps.

II. NOTATION, CONVENTIONS, AND DIMENSION ANALYSIS

We denote by D the cosmological distance to a source, and by $\ell(D)$ the effective line-of-sight propagation length over which foam fluctuations accumulate; for sources at moderate redshifts, $\ell(D) \approx D$.

A. Foam fluctuation density and ray-dependent path length

We model the photon path as propagating along a mean axis, with a transverse coordinate $\mathbf{x}_\perp$ perpendicular to that axis. We represent an individual ray by a small direction offset $\boldsymbol{\theta}$ (e.g. far-field small-angle approximation) such that the ray's transverse position at distance s along the line of sight is approximately

$$\mathbf{x}_\perp(s;\boldsymbol{\theta}) \simeq s\,\boldsymbol{\theta}.$$

We model foam fluctuations by a *dimensionless* fluctuation density $\eta(s,\mathbf{x}_\perp,t)$, where s is distance along the mean propagation direction and t is time. The total path-length fluctuation accumulated to a photon propagating along ray $\boldsymbol{\theta}$ is

$$\Delta\ell^{(D)}_{\rm tot}(t;\boldsymbol{\theta}) \;\equiv\; \int_0^{\ell(D)} \eta(s,\,\mathbf{x}_\perp(s;\boldsymbol{\theta}),\,t)\;ds. \tag{2}$$

This quantity has dimension of length. The foam-induced phase at the detector is

$$\varphi^{(D)}(t;\boldsymbol{\theta}) \;\equiv\; \frac{2\pi}{\lambda}\,\Delta\ell^{(D)}_{\rm tot}(t;\boldsymbol{\theta}), \tag{3}$$

which is dimensionless.

B. Arrival-time jitter

Arrival-time jitter associated with ray $\boldsymbol{\theta}$ is given by

$$\delta t(t;\boldsymbol{\theta}) \;=\; \frac{1}{c}\Delta\ell^{(D)}_{\rm tot}(t;\boldsymbol{\theta}). \tag{4}$$

This depends on the *ray label* because the transverse foam sampling differs across rays.

C. Temporal and transverse correlations

We encode foam statistics through the following two ingredients:

- **Temporal decorrelation** of path aggregates along a single ray:

$$C_\alpha(\tau;D) \;\equiv\; \frac{\left\langle \Delta\ell^{(D)}_{\rm tot}(t;\boldsymbol{\theta})\,\Delta\ell^{(D)}_{\rm tot}(t+\tau;\boldsymbol{\theta})\right\rangle}{\sigma^2_\ell(D)}, \qquad C_\alpha(0;D)=1, \tag{5}$$

  where $\sigma^2_\ell(D)$ is the single-ray variance (independent of $\boldsymbol{\theta}$ under statistical homogeneity).

- **Transverse decorrelation** between different rays: we introduce a transverse correlation function $\Phi(u)$ with $\Phi(0)=1$ and $\Phi(u)\to 0$ for large u, controlled by a transverse correlation length $r_\perp$.

To connect rays, we define a typical transverse separation between two rays. In the far-field small-angle limit, a standard estimate is

$$b(\boldsymbol{\theta},\boldsymbol{\theta}') \simeq \ell(D)\,|\boldsymbol{\theta}-\boldsymbol{\theta}'|. \tag{6}$$

D. Path-aggregate Ng–Perlman scaling

The central ansatz is that the single-ray variance inherits the Ng–Perlman scaling:

$$\sigma_\ell^2(D) \;=\; A_\alpha\,[\ell(D)]^{2(1-\alpha)}\,\ell_{\rm P}^{2\alpha}, \tag{7}$$

where A_α is dimensionless and order unity (up to mild dependence on α and detailed correlation structure, summarized later).

Dimensional consistency: $\big[\ell(D)\big]^{2(1-\alpha)}\,\ell_{\rm P}^{2\alpha}$ has dimension length^2.

E. Foam two-ray covariance closure

We assume that the covariance of path-length aggregates between two rays factorizes into temporal and transverse pieces:

$$\Big\langle \Delta\ell_{\rm tot}^{(D)}(t;\boldsymbol{\theta})\,\Delta\ell_{\rm tot}^{(D)}(t+\tau;\boldsymbol{\theta}')\Big\rangle \;=\; \sigma_\ell^2(D)\,C_\alpha(\tau;D)\,\Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right). \tag{8}$$

This closure satisfies the correct limits: at $\tau=0$ and $\boldsymbol{\theta}=\boldsymbol{\theta}'$ we recover $\sigma_\ell^2(D)$.

Finally, we maintain statistical independence between intrinsic source fluctuations and foam fluctuations.

III. FOAM-INDUCED TEMPORAL AND TRANSVERSE PHASE/JITTER STATISTICS

A. Two-time phase variance (generalized)

The two-time phase difference relevant for temporal coherence for ray $\boldsymbol{\theta}$ is

$$\Delta\varphi^{(D)}(\tau;\boldsymbol{\theta}) \equiv \varphi^{(D)}(t+\tau;\boldsymbol{\theta}) - \varphi^{(D)}(t;\boldsymbol{\theta}). \tag{9}$$

For the present purpose (arrival-time jitter), we emphasize timing rather than phase. Still, the same temporal covariance enters the variance:

$$\Big\langle \big[\Delta\varphi^{(D)}(\tau;\boldsymbol{\theta})\big]^2\Big\rangle = 2\left(\frac{2\pi}{\lambda}\right)^2 \sigma_\ell^2(D)\,[1-C_\alpha(\tau;D)]. \tag{10}$$

For a *two-ray* difference one would similarly obtain factors of Φ; we do not require the phase variance explicitly below because $g^{(2)}$ is sensitive to arrival-time jitter.

B. Differential arrival-time jitter between rays

For HBT, photons detected at times separated by τ may originate from different rays within the detected spatial mode. Define the differential arrival-time increment between rays $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$ as

$$\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau) \;\equiv\; \delta t(t+\tau;\boldsymbol{\theta}) - \delta t(t;\boldsymbol{\theta}'). \tag{11}$$

Using Eq. (4), Eq. (8), and stationarity, its variance is

$$\begin{aligned}
\sigma_\Delta^2(\tau;\boldsymbol{\theta},\boldsymbol{\theta}') &\equiv \Big\langle \big[\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau)\big]^2\Big\rangle \\
&= \frac{1}{c^2}\Big[\langle\Delta\ell(t+\tau;\boldsymbol{\theta})^2\rangle + \langle\Delta\ell(t;\boldsymbol{\theta}')^2\rangle - 2\langle\Delta\ell(t+\tau;\boldsymbol{\theta})\Delta\ell(t;\boldsymbol{\theta}')\rangle\Big] \\
&= \frac{2\sigma_\ell^2(D)}{c^2}\Big[1 - C_\alpha(\tau;D)\,\Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right)\Big].
\end{aligned} \tag{12}$$

For $\boldsymbol{\theta}=\boldsymbol{\theta}'$, $\Phi(0)=1$ and one recovers the earlier same-ray result:

$$\sigma_\Delta^2(\tau;\boldsymbol{\theta},\boldsymbol{\theta}) = \frac{2\sigma_\ell^2(D)}{c^2}\,[1-C_\alpha(\tau;D)].$$

C. Gaussianity

As in the 1D formalism, we assume that path-aggregate fluctuations are sums over many weakly correlated foam cells. For cosmological distances this motivates a Gaussian approximation for the differential jitter:

$$\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau) \sim \mathcal{N}\big(0, \sigma_\Delta^2(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')\big)\,. \tag{13}$$

We will therefore use a Gaussian jitter kernel for each ray pair.

IV. FROM FOAM-INDUCED JITTER TO THE HBT OBSERVABLE

A. Source and field statistics

We consider a quasi-monochromatic chaotic (thermal) source with central angular frequency $\omega_0 = 2\pi c/\lambda$. In the absence of foam, the detected analytic signal is modeled as a zero-mean circular complex Gaussian random process.

We include an extended source by allowing contributions from rays labeled by $\boldsymbol{\theta}$ (within the angular region coupled into the detection mode). We treat the foam as imprinting ray-dependent arrival-time jitter, not only phases.

A minimal operational picture for photon-counting HBT is: the intrinsic photon arrival times are shifted by a random $\delta t(t;\boldsymbol{\theta})$ that depends on which ray the photon propagated along.

B. Siegert relation and the role of arrival-time jitter

For chaotic Gaussian fields, the Siegert relation holds:

$$g^{(2)}(\tau) \;=\; 1 + \big|g^{(1)}(\tau)\big|^2. \tag{14}$$

In the earlier 1D approach, a strictly pure-phase, single-mode foam left $g^{(2)}(\tau)$ unchanged because $|E(t)|^2$ is insensitive to phase. That cancellation still holds when the foam only modulates the phase without producing timing (or without coupling phase to intensity through wavefront distortion). In contrast, arrival-time jitter does alter photon-counting coincidences: it displaces detected photons along the time axis and smears the coincidence peak.

Therefore, our operative mechanism for temporal HBT constraints is the arrival-time jitter, generalized here to include transverse decorrelation between rays.

V. ANGULARLY AVERAGED CONVOLUTION FORMULA FOR $g^{(2)}_{\rm obs}(\tau)$

A. Jitter kernels and extended-source averaging

Let $g^{(2)}_{\rm src}(\tau)$ denote the intrinsic second-order correlation function of the source (in the absence of foam). In the presence of arrival-time jitter, the measured coincidence peak at lag τ is built from photon pairs whose true time separation differs from the nominal lag by a differential jitter $\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau)$.

For a fixed pair of rays $(\boldsymbol{\theta},\boldsymbol{\theta}')$, the jitter PDF is Gaussian:

$$p_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\Delta;\tau) = \frac{1}{\sqrt{2\pi}\,\sigma_\Delta(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')}\exp\!\left(-\frac{\Delta^2}{2\sigma_\Delta^2(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')}\right). \tag{15}$$

The observed $g^{(2)}_{\rm obs}$ becomes a mixture over ray pairs, weighted by an angular acceptance/source weighting function $W(\boldsymbol{\theta},\boldsymbol{\theta}')$ that encodes: (i) source angular brightness distribution, (ii) telescope point-spread/collection mode, and (iii) how intensity at the detector samples the field from different angles. We require only normalization $\iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\, W(\boldsymbol{\theta},\boldsymbol{\theta}') = 1$ for convenience.

The general angularly averaged convolution is then

$$\boxed{g^{(2)}_{\rm obs}(\tau) - 1 = \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; W(\boldsymbol{\theta},\boldsymbol{\theta}') \int_{-\infty}^{\infty} p_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\Delta;\tau)\left[g^{(2)}_{\rm src}(\tau+\Delta) - 1\right] d\Delta.} \tag{16}$$

B. Effective transverse coherence approximation

In many practical cases one can define an effective transverse-coherence factor by averaging Φ over the angular region that significantly contributes to the detected intensity correlations:

$$\Phi_{\rm eff}(\tau) \;\equiv\; \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; W(\boldsymbol{\theta},\boldsymbol{\theta}')\, \Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right). \tag{17}$$

Assuming the mixture is approximately represented by a single Gaussian kernel with variance

$$\sigma^2_{\tau,\rm eff}(D;\tau) \;\equiv\; \frac{2\sigma^2_\ell(D)}{c^2}\left[1 - C_\alpha(\tau;D)\,\Phi_{\rm eff}\right], \tag{18}$$

Eq. (16) reduces to the earlier 1D structure with $\sigma_\tau \to \sigma_{\tau,\rm eff}$.

C. Point-like vs extended limits controlled by Φ

Equation (12) makes the regimes explicit:

- **Point-like (or fully correlated) regime:** if transverse foam coherence is large across the angular support of W, then $\Phi(b/r_\perp) \approx 1$, hence $\Phi_{\rm eff} \approx 1$ and

  $$\sigma^2_{\tau,\rm eff}(D;\tau) \approx \frac{2\sigma^2_\ell(D)}{c^2}\,[1 - C_\alpha(\tau;D)],$$

  recovering the earlier 1D same-ray result.

- **Extended (incoherent transverse) regime:** if transverse coherence is small, then for typical ray separations $\Phi(b/r_\perp) \approx 0$ and $\Phi_{\rm eff} \approx 0$, giving

  $$\sigma^2_{\tau,\rm eff}(D;\tau) \approx \frac{2\sigma^2_\ell(D)}{c^2},$$

  i.e. maximal broadening relative to the correlated case.

D. Lorentzian source example

For a quasi-monochromatic chaotic source with coherence time τ_c, take

$$g^{(2)}_{\rm src}(\tau) - 1 = \exp\!\left(-\frac{|\tau|}{\tau_c}\right). \tag{19}$$

Under the effective Gaussian approximation, the convolution yields the same functional form as in the 1D case upon replacing σ_τ by $\sigma_{\tau,\rm eff}$. Define $\sigma_{\tau,\rm eff} \equiv \sigma_\tau$ for notational brevity. Then

$$g^{(2)}_{\rm obs}(\tau) - 1 = \exp\!\left(\frac{\sigma^2_\tau}{2\tau_c^2} - \frac{|\tau|}{\tau_c}\right) \operatorname{erfc}\!\left(\frac{\sigma_\tau/\tau_c - |\tau|/\sigma_\tau}{\sqrt{2}}\right) \cdot \frac{1}{2} e^{|\tau|/\tau_c}. \tag{20}$$

Two limiting regimes are:

$$\sigma_\tau \ll \tau_c: \quad g^{(2)}_{\rm obs}(\tau) - 1 \approx \exp\!\left(-\frac{|\tau|}{\tau_c}\right), \tag{21}$$

$$\sigma_\tau \gg \tau_c: \quad g^{(2)}_{\rm obs}(\tau) - 1 \approx \frac{\tau_c}{\sqrt{2\pi}\,\sigma_\tau} \exp\!\left(-\frac{\tau^2}{2\sigma^2_\tau}\right). \tag{22}$$

Finally, note that $\sigma_{\tau,\rm eff}(\tau = 0) = 0$ because $C_\alpha(0;D) = 1$, ensuring $g^{(2)}_{\rm obs}(0) = 2$ in the idealized Gaussian single-mode detection limit.

VI. PARAMETERIZED NUMERICAL ILLUSTRATION

A. Representative parameters

We evaluate the foam-induced rms arrival-time jitter scale

$$\sigma^2_{\tau,\mathrm{eff}}(D;\tau) = \frac{2\sigma^2_\ell(D)}{c^2}\Big[1 - C_\alpha(\tau;D)\,\Phi_{\mathrm{eff}}\Big], \qquad \sigma^2_\ell(D) = A_\alpha[\ell(D)]^{2(1-\alpha)}\ell_{\mathrm{P}}^{2\alpha}.$$

For definiteness, set $\ell(D) \approx D$, $A_\alpha = 1$, and take maximal decorrelation in time $1 - C_\alpha(\tau;D) \sim 1$ (i.e. consider the large-lag regime where the temporal correlation has decayed). For the source coherence time, take $\tau_c \sim \lambda^2/(c\Delta\lambda)$ with a narrow optical filter: $\Delta\lambda/\lambda \sim 10^{-3}$ at $\lambda = 500$ nm, giving $\tau_c \approx 0.8$ ps as in the earlier 1D illustration.

We set $D = 1$ Gpc $\approx 3.09 \times 10^{25}$ m.

B. Model foam scaling for $\alpha \in \{1, 2/3, 1/2\}$

Assuming $\Phi_{\mathrm{eff}} = 1$ corresponds to the point-like correlated regime and $\Phi_{\mathrm{eff}} = 0$ corresponds to the maximal transverse decoherence regime, the characteristic jitter scale in the large-lag limit becomes

$$\sigma_{\tau,\mathrm{eff}} \sim \frac{\sqrt{2}\,\sigma_\ell(D)}{c} \quad (\text{for } \Phi_{\mathrm{eff}} \approx 0 \text{ or } 1 - C_\alpha \approx 1).$$

Using the same scalings as the 1D calculation yields:

$$\alpha = 1: \quad \sigma_\ell = \ell_{\mathrm{P}} = 1.616 \times 10^{-35}\,\mathrm{m} \quad \Rightarrow \quad \sigma_{\tau,\mathrm{eff}} \approx 7.6 \times 10^{-44}\,\mathrm{s}, \tag{23}$$

$$\alpha = 2/3: \quad \sigma_\ell \approx 5.7 \times 10^{-15}\,\mathrm{m} \quad \Rightarrow \quad \sigma_{\tau,\mathrm{eff}} \approx 2.7 \times 10^{-23}\,\mathrm{s}, \tag{24}$$

$$\alpha = 1/2: \quad \sigma_\ell \approx 2.2 \times 10^{-5}\,\mathrm{m} \quad \Rightarrow \quad \sigma_{\tau,\mathrm{eff}} \approx 1.0 \times 10^{-13}\,\mathrm{s} = 0.10\,\mathrm{ps}. \tag{25}$$

Thus the random-walk model can yield σ_τ comparable to τ_c, while the holographic and conservative models remain far below detectability, even before including the additional potential suppression from transverse decoherence (which, in our framework, only *increases* the effective jitter by decreasing Φ_{eff}).

C. A transverse-coherence example

To visualize the dependence on extended-source physics, adopt a concrete transverse correlation form:

$$\Phi(u) = \exp\left(-\frac{u^2}{2}\right), \qquad u = \frac{b}{r_\perp}.$$

Then Φ_{eff} is determined by the typical angular separation contributing to W: with $b \simeq \ell(D)|\boldsymbol{\theta} - \boldsymbol{\theta}'|$, the criterion $u \ll 1$ corresponds to $r_\perp \gg \ell(D)\Delta\theta$ (point-like in the foam sense), while $u \gg 1$ gives $\Phi_{\mathrm{eff}} \approx 0$ (maximal transverse decorrelation).

For illustration, one may parameterize Φ_{eff} by a single number $\Phi_{\mathrm{eff}} \in [0,1]$ and note:

$$\sigma^2_{\tau,\mathrm{eff}}(D;\tau) = \frac{2A_\alpha}{c^2}\,[\ell(D)]^{2(1-\alpha)}\ell_{\mathrm{P}}^{2\alpha}\Big[1 - C_\alpha(\tau;D)\Phi_{\mathrm{eff}}\Big].$$

In the correlated regime $\Phi_{\mathrm{eff}} \to 1$ one recovers the earlier 1D prediction, while in the incoherent transverse regime $\Phi_{\mathrm{eff}} \to 0$ the effective jitter increases.

VII. DISCUSSION

A. Pure-phase cancellation vs arrival-time jitter

The original motivation for temporal HBT foam constraints relies on the fact that intensity correlations are insensitive to a *strictly* pure phase modulation of a single-mode field: if the foam affects only the phase and does not produce

arrival-time jitter (and does not couple phase to intensity through wavefront distortion), then the predicted $g^{(2)}(\tau)$ is unchanged. In this work we explicitly include the arrival-time jitter mechanism. Transverse foam fluctuations only matter insofar as they make the jitter ray-dependent and reduce the transverse coherence factor Φ between different rays contributing to the detected correlation.

B. Extended sources and the transverse coherence criterion

The key new result is that the relevant decoherence channel is controlled by

$$1 - C_\alpha(\tau; D)\,\Phi\left(\frac{b}{r_\perp}\right).$$

Thus, even if the temporal foam correlation length yields significant time decorrelation, the foam's ability to suppress the HBT bunching peak depends on whether the transverse coherence of the foam is large enough to keep different rays correlated.

In particular:

- If $r_\perp \gg \ell(D)\Delta\theta$, then $\Phi \approx 1$ over the angular support and the extended source reduces to an effective point-source foam response.
- If $r_\perp \ll \ell(D)\Delta\theta$, then $\Phi \approx 0$ for typical ray separations and the extended-source case experiences increased effective jitter, further broadening and suppressing $g^{(2)}(\tau) - 1$.

C. Limitations

This formulation is phenomenological in two places: (i) the Ng–Perlman scaling for single-ray variance through A_α, and (ii) the factorized temporal/transverse closure Eq. (8). Future work should derive Φ and its relation to microscopic foam correlation properties. Additionally, the angular weighting function $W(\boldsymbol{\theta}, \boldsymbol{\theta}')$ depends on the detection mode and source brightness distribution; a fully realistic prediction requires specifying observational configuration (e.g. coupling into a single-mode fiber vs imaging camera, telescope PSF, and source angular structure).

D. Next steps

To complete an observational program, one should: (i) build a model of W for a given instrument and source, (ii) adopt or derive plausible foam transverse correlation models $\Phi(u)$, (iii) compute $g^{(2)}_{\rm obs}(\tau)$ in either the full mixture form Eq. (16) or under the effective Gaussian approximation Eq. (18), and (iv) compare predicted bunching suppression/broadening against measured temporal HBT peaks.

VIII. CONCLUSIONS

We presented a dimensionally consistent formalism for temporal HBT constraints on spacetime foam that incorporates both temporal and transverse foam fluctuations. The core mechanism is photon arrival-time jitter generated by accumulated path-length fluctuations. We derived the variance of the differential arrival-time jitter between two rays:

$$\sigma^2_\Delta(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}') = \frac{2\sigma^2_\ell(D)}{c^2}\left[1 - C_\alpha(\tau; D)\,\Phi\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right)\right],$$

with $\sigma^2_\ell(D) = A_\alpha[\ell(D)]^{2(1-\alpha)}\ell_{\rm P}^{2\alpha}$. The observed correlation is an angularly averaged convolution of the intrinsic source correlation with the jitter kernel:

$$g^{(2)}_{\rm obs}(\tau) - 1 = \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; W(\boldsymbol{\theta}, \boldsymbol{\theta}') \int d\Delta\, p_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\Delta; \tau)\left[g^{(2)}_{\rm src}(\tau + \Delta) - 1\right].$$

In the limit of large transverse foam coherence across the source ($\Phi \approx 1$), the prediction reduces to the earlier line-of-sight (1D) result. When transverse foam coherence is small ($\Phi \approx 0$), extended sources experience enhanced effective jitter, broadening and suppressing the HBT bunching peak. Pure-phase, strictly single-mode foam leaves $g^{(2)}(\tau)$ unchanged; observable effects require arrival-time jitter and/or phase-to-amplitude coupling mechanisms.

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Appendix A: Conditions for Ng–Perlman Scaling of the Path-Aggregate Variance

Here we discuss conditions under which the path-aggregate variance

$$\sigma_\ell^2(D) = \int_0^{\ell(D)} ds \int_0^{\ell(D)} ds'\, S(s,s')$$

inherits the Ng–Perlman single-point scaling

$$\sigma_\ell^2(D) = A_\alpha [\ell(D)]^{2(1-\alpha)} \ell_{\mathrm{P}}^{2\alpha}.$$

In this work we treat A_α phenomenologically; for completeness we summarize key correlation-structure considerations.

1. Short-range correlations (white-noise limit)

If the foam fluctuation density $\eta(s,t)$ has a finite correlation length ξ along the line of sight, and

$$S(s,s') = \sigma_0^2 f\left(\frac{|s-s'|}{\xi}\right), \qquad f(0)=1, \qquad \int_0^\infty f(u)\,du = 1,$$

then for $\ell(D) \gg \xi$,

$$\sigma_\ell^2(D) \approx \sigma_0^2\, \xi\, \ell(D),$$

which corresponds to random-walk growth $\alpha = 1/2$ when $\sigma_0^2 \xi \sim A_{1/2}\ell_{\mathrm{P}}$.

2. Long-range correlations and other α values

For correlations of the power-law form $S(u) \propto u^{-\beta}$ with $0 < \beta < 1$, the double integral scales as $\sigma_\ell^2 \propto \ell(D)^{2-\beta}$. Matching $2(1-\alpha) = 2-\beta$ gives $\beta = 2\alpha$. Thus, for sub-random-walk $\alpha > 1/2$ one requires long-range/anticorrelated structures such that the effective integral scaling decreases relative to the $\alpha = 1/2$ baseline.

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